

RESEARCH AND DEVELOPMENT PROPOSAL

(Submitted for Supervisory Review and Institutional Consideration)

SETU

Structural Evaluation and Thickness Underwriting

A Reliability-Based Artificial Intelligence Framework for Residual-Life Prediction and Maintenance Prioritisation of Concrete Pavements

Particulars	Details
Broad Subject Area	Civil Engineering — Transportation / Pavement Engineering
Sub-Area	Rigid Pavement Design, Reliability Analysis, Applied Machine Learning, Infrastructure Asset Management
Proposed by	[Name of Investigator]
Department	Department of Civil Engineering
Institution	[Name of Institution]
Supervisor	[Name and Designation of Supervisor]
Proposed Duration	24 months (Phase I: 12 months; Phase II: 12 months)
Nature of Proposal	Applied research with translational and commercialisation potential
Date of Submission	_____

Fields marked [] are to be completed prior to formal institutional submission.

1. EXECUTIVE SUMMARY

Concrete (rigid) pavements are designed once, at the planning stage, using single representative values for subgrade support, concrete strength, temperature differential and projected traffic. In reality each of these quantities varies continuously along the alignment of a highway. The prevailing practice accommodates this variability by adopting a uniform design thickness together with a blanket margin of safety applied along the entire stretch.

This produces two simultaneous and opposite inefficiencies. Where ground conditions are more favourable than assumed, concrete is provided in excess of requirement, at avoidable cost and avoidable embodied carbon. Where conditions are less favourable, distress develops earlier than the design life, and the owning authority has no means of identifying which segments are at risk until visible failure occurs.

The proposed work develops SETU, a reliability-based computational framework that treats the governing design parameters as statistical distributions rather than deterministic values, propagates that uncertainty through the cumulative fatigue damage formulation prescribed in IRC:58, and thereby produces a thickness requirement that varies along the alignment at a stated level of reliability, together with a continuously updated estimate of residual life and distress risk for each segment of an in-service pavement. A machine-learning layer is superposed on the mechanistic core to correct systematic deviation between predicted and observed distress, without displacing the codal formulation.

The framework requires no additional field instrumentation. It operates upon records that highway projects are already required to generate. A verified computational core has been implemented and tested, and is described in Section 8.

2. INTRODUCTION AND BACKGROUND

India has undertaken the largest programme of concrete road construction in its history, and a substantial proportion of that network is now approaching the age at which fatigue-related distress characteristically begins to appear. Concurrently, the institutional framework for highway delivery has shifted towards concession-based models, in which a private entity assumes responsibility for pavement condition over a period of fifteen to thirty years. For such an entity the timing and location of maintenance intervention is a direct financial exposure. There exists, for the first time, an institutional stakeholder with a quantified commercial incentive to predict pavement deterioration in advance.

The design of plain jointed rigid pavements in India is governed by IRC:58, which prescribes a mechanistic-empirical procedure founded upon the cumulative fatigue damage concept. The formulation is sound. The limitation addressed in this proposal does not lie in the mechanics, but in the manner of their application: the computation is performed once, with deterministic inputs, and is never subsequently revisited against the conditions actually realised on site.

3. STATEMENT OF THE PROBLEM

Design parameter	Treatment in practice	Field reality	Typical CoV
Modulus of subgrade reaction, k	Single value from limited plate load tests or CBR correlation	Varies continuously along the alignment with soil type, moisture and compaction	0.20 – 0.35
Flexural strength (modulus of rupture)	Single specified value	Varies between pours, batches and seasons	0.10 – 0.15
Elastic modulus, E	Single value or correlation	Varies with mix, aggregate and maturity	0.10 – 0.15
Slab thickness, h	Nominal design value	Construction tolerance and sub-base irregularity	0.02 – 0.05
Traffic (commercial vehicles)	Projected figure from a census	Projection error, overloading and diversion	0.15 – 0.30

Table 3.1 — Divergence between design assumption and field reality. Coefficients of variation are indicative.

The central question that the industry is presently unable to answer may be stated concisely:

“Which segment of this pavement will fail first, when will it fail, and what is the financial consequence of deferring intervention?”

The data required to answer this question is, in substantial part, already generated and archived in the ordinary course of highway construction and operation. It is not analysed in an integrated manner, and it is not related back to the mechanistic damage model from which the design was originally derived.

4. REVIEW OF LITERATURE AND STATE OF THE ART

4.1 Mechanistic basis

Westergaard (1926) established the closed-form solution for stresses in a concrete slab on a dense-liquid foundation for interior, edge and corner loading; Ioannides, Thompson and Barenberg (1985) reconsidered and corrected these solutions. Bradbury (1938) established the coefficient governing restraint to thermal curling. The fatigue behaviour of concrete under repeated flexural stress is characterised by the stress ratio, and the Portland Cement Association (1984) formalised the relationship between stress ratio and allowable repetitions. Miner (1945) provides the linear damage summation hypothesis. IRC:58 adopts this framework for Indian conditions.

4.2 Reliability in pavement design

The treatment of pavement design parameters as random variables is well established internationally, and reliability concepts appear in the AASHTO framework through a reliability level and an overall standard deviation. However, the reliability factor in such formulations is applied as a global multiplier and does not resolve variability at the level of individual segments of a specific alignment.

4.3 Machine learning in pavement engineering

Machine learning has been applied extensively to pavement condition prediction, principally by correlating condition indices with age, traffic and climate. Two limitations recur. First, purely data-driven models require extensive historical condition records, seldom available at the necessary resolution in the Indian context. Second, such models do not admit interpretation in terms of the governing codal provisions, and consequently cannot readily be relied upon by an engineer who must accept professional responsibility for the recommendation.

4.4 Pavement management systems

Established pavement management systems operate principally at network level and are oriented towards flexible pavements, employing statistical condition indices. They are not generally constructed upon slab-level mechanistic damage accumulation, and their calibration to Indian rigid pavements under IRC provisions has received limited attention.

5. RESEARCH GAP IDENTIFIED

- **Deterministic application of a probabilistic phenomenon.** The cumulative fatigue damage formulation is applied once, as a design check, and not forward in time as a predictive instrument.
- **Absence of segment-level resolution.** Reliability is introduced through global factors rather than segment-resolved analysis.
- **Disjunction between mechanistic and data-driven methods.** A hybrid formulation in which codal mechanics provide the prediction and machine learning corrects only the residual has not been established for Indian rigid pavements.
- **Non-utilisation of existing construction records.** Pour registers, cube results and subgrade data are archived for contractual compliance and never related to the mechanistic model.

6. OBJECTIVES OF THE PROPOSED WORK

Objective 1: To formulate a probabilistic extension of the IRC:58 cumulative fatigue damage procedure, in which subgrade reaction, flexural strength, elastic modulus, slab thickness and traffic volume are represented as statistical distributions, and to establish by Monte Carlo simulation the resulting distribution of cumulative fatigue damage and the associated probability of failure.

Objective 2: To develop a reliability-based thickness determination procedure, yielding the minimum slab thickness required to achieve a stated reliability for a given segment, quantifying the material saving achievable on favourable segments and the additional provision required on unfavourable segments.

Objective 3: To incorporate thermal warping and both cracking mechanisms, so that bottom-up cracking under the daytime positive temperature differential and top-down cracking under the reversed night-time gradient are evaluated, and the governing mechanism identified.

Objective 4: To establish a residual-life and distress-risk estimation procedure, by applying the damage accumulation model forward in time using as-built material and construction data together with observed traffic and climate.

Objective 5: To develop a hybrid predictive formulation, in which a machine-learning model is trained upon the residual between mechanistically predicted and field-observed distress, accommodating local effects without displacing the codal basis of the prediction.

Objective 6: To formulate a maintenance prioritisation procedure, ranking segments by the expected cost of deferred intervention.

Objective 7: To validate the framework, by retrospective application to existing concrete pavement sections of known condition and construction history.

7. METHODOLOGY

The framework is organised in four computational layers. The arrangement is deliberate: the deterministic codal formulation is retained as the foundation, and each successive layer adds information without displacing the layer beneath it. At zero input variability the framework reduces exactly to the conventional IRC:58 computation, and this property has been verified by automated test.

Layer	Function	Basis	Output
Layer 1** Mechanistic core	Flexural stress from load and thermal warping; stress ratio; allowable repetitions; cumulative fatigue damage	Westergaard; Bradbury; IRC:58 fatigue formulation; Miner	Deterministic damage state, clause-traceable
Layer 2** Probabilistic propagation	Monte Carlo simulation over distributions of k, MR, E, h and traffic	Reliability theory; lognormal representation	Damage distribution; probability of failure
Layer 3** ML residual correction	Regression upon the residual between predicted and observed distress	Gaussian Process regression	Corrected forecast with confidence interval
Layer 4** Decision engine	Cost of present intervention against probability-weighted cost of deferral	Expected-value decision analysis	Ranked maintenance schedule

Table 7.1 — Layered architecture of the SETU framework.

7.1 Rationale for the hybrid formulation

- **Professional accountability.** A prediction produced by an uninterpretable model cannot be relied upon by an engineer who must accept professional responsibility for it.
- **Data efficiency.** Where the mechanistic model supplies the principal signal, the learned component need only account for a small residual, and useful accuracy is attainable with datasets of a few hundred observations.
- **Regulatory admissibility.** Output that cites the governing clause is admissible in project documentation; output that cannot be so traced is not.

7.2 Data sources

The framework operates upon records already generated in the ordinary course of construction and operation, and requires no instrumentation: subgrade and sub-base investigation records; concrete pour registers and cube or beam test results; batching plant records; traffic census and toll plaza classification data; meteorological records; and periodic condition surveys.

7.3 Validation strategy

- **Retrospective validation** against existing sections of known age, construction record and documented condition.
- **Accelerated segments** — heavily trafficked corridors on which damage accumulates more rapidly, shortening the interval for prospective confirmation.
- **Analytical verification** — confirmation that the framework reproduces the deterministic IRC:58 result at zero variability and preserves all monotonic physical relationships.

8. PRELIMINARY WORK ALREADY COMPLETED

A functioning computational core has been implemented and verified prior to the submission of this proposal. The work is not conceptual.

Component	Description	Status
Analytical stress module	Westergaard interior, edge and corner stress and deflection; radius of relative stiffness; equivalent radius of resisting section	Implemented and verified
Thermal warping module	Bradbury coefficient by continuous fitted function; edge and interior warping stress; bottom-up and top-down cracking checks	Implemented and verified
Fatigue and damage module	Stress ratio; allowable repetitions across the three fatigue branches; Miner summation; residual life with traffic growth	Implemented and verified
Reliability module	Monte Carlo propagation; probability of failure; reliability index; bisection search for reliability-based thickness	Implemented and verified
Public computation tool	Browser-based calculator with complete calculation trace, verified against the reference implementation	Implemented and verified
Verification test suite	56 automated tests encoding physical invariants, fit accuracy, and reduction to the deterministic codal result	56 of 56 passing

Table 8.1 — Status of the implemented computational core.

8.1 Reliability-based thickness

Two segments of a single notional corridor, identical in every respect except subgrade quality. A conventional deterministic design would specify one uniform thickness for both.

Modulus of subgrade reaction, k (MPa/mm)	Thickness for 90% reliability (mm)	Departure from uniform design
0.200 (very good)	285	15 mm less
0.150 (good)	295	5 mm less
0.110 (fair)	300	reference
0.080 (poor)	305	5 mm more required
0.045 (very poor)	320	20 mm more required

Table 8.2 — Reliability-based thickness requirement as a function of subgrade quality, at a target reliability of 90 per cent.

Computed using the implemented module.

The spread of 35 mm across the same corridor is the quantity a uniform design necessarily conceals. On favourable segments the framework identifies material that may safely be omitted; on unfavourable segments it identifies provision required to attain the same reliability. Both results proceed from a single computation.

8.2 Significance of thermal warping

The implemented thermal module demonstrates that warping stress frequently governs. For a 300 mm slab under a 15 tonne axle with a daytime differential of 16.8 °C:

Analysis	Stress (MPa)	Stress ratio	Assessment
Load only	2.055	0.457	below the endurance limit; apparently safe
Load and warping combined (BUC governs)	3.625	0.806	substantially above the endurance limit

Table 8.3 — Effect of thermal warping on the governing stress ratio.

Omitting thermal warping understates the stress ratio by approximately 43 per cent. Because the fatigue relationship is steeply non-linear in the stress ratio, this becomes an error of orders of magnitude in the allowable number of repetitions. Warping is therefore not a refinement but frequently the governing consideration.

These figures are illustrative and employ indicative coefficients of variation. Project-specific values derived from the client's own test records are to be substituted before any design application.

9. EXPECTED OUTCOMES AND DELIVERABLES

Sl.	Deliverable	Nature
1	A validated probabilistic extension of the IRC:58 cumulative fatigue damage procedure	Methodological
2	A reliability-based thickness determination procedure with quantified material and carbon saving	Design methodology
3	A residual-life and distress-risk estimation procedure	Asset management
4	A hybrid mechanistic-machine-learning model with calibrated uncertainty	Computational
5	A maintenance prioritisation procedure with financial consequence	Decision support
6	A verified open computational implementation with test suite	Software
7	Publications and professional presentations	Dissemination
8	A demonstration study on an identified corridor	Field validation

10. WORK PLAN AND TIME SCHEDULE

Phase	Activity	Months	Milestone
I	Literature consolidation; variability statistics for Indian materials and subgrades	1 – 3	Parameter database
I	Extension of the mechanistic core: temperature differential regimes and erosion criterion	3 – 6	Extended Layer 1
I	Development and verification of the probabilistic layer	5 – 9	Validated Layer 2
I	Corridor identification; acquisition of construction and condition records	7 – 12	Field dataset
II	Retrospective validation against documented distress	12 – 16	Accuracy established
II	Machine-learning residual layer	14 – 19	Validated Layer 3
II	Decision engine and maintenance prioritisation	18 – 21	Layer 4 complete
II	Demonstration study, documentation and dissemination	20 – 24	Publications; final report

11. BUDGET ESTIMATE

The proposal is deliberately structured to require minimal capital outlay. No field instrumentation, proprietary software licence or high-performance computing facility is necessary; the computational methods execute on ordinary desktop hardware using open-source components.

Head	Year 1 (Rs.)	Year 2 (Rs.)	Total (Rs.)
Manpower (Junior Research Fellow, if sanctioned)	3,72,000	3,72,000	7,44,000
Travel and field visits	60,000	80,000	1,40,000
Consumables and data acquisition	40,000	40,000	80,000

Head	Year 1 (Rs.)	Year 2 (Rs.)	Total (Rs.)
Contingency	25,000	25,000	50,000
Publication and dissemination	20,000	60,000	80,000
Total	5,17,000	5,77,000	10,94,000

The budget is indicative and may be reduced substantially if the work is undertaken without a sanctioned fellowship. The computational component carries no cost.

12. UTILISATION OF RESULTS AND COMMERCIALISATION POTENTIAL

12.1 Academic and professional utilisation

- Contribution towards the eventual incorporation of reliability-based provisions in Indian rigid pavement design practice.
- Provision of a verified open computational implementation for researchers and practitioners.
- Establishment of a database of parameter variability for Indian materials and subgrade conditions.

12.2 Institutional utilisation

- Highway authorities may allocate constrained maintenance budgets on an objective and auditable basis in place of uniform or reactive allocation.
- Concessionaires may forecast maintenance liability and schedule intervention before distress becomes visible.
- Design consultants may offer reliability-based design as a differentiated professional service.

12.3 Commercialisation pathway

Stage	Offering	Basis
1	Open reliability-based design calculator, freely available	Dissemination and professional adoption
2	Residual-life and maintenance prioritisation study for a specified corridor	Professional consultancy
3	Continuously updated condition and risk platform	Subscription
4	Network-level prioritisation across a portfolio	Institutional engagement

The economic proposition rests upon a well-documented characteristic of pavement maintenance, namely that the cost of intervention increases sharply, and non-linearly, once distress has become manifest. A framework that redirects even a modest proportion of an existing maintenance budget from segments not requiring intervention to segments that do will recover its cost within a single maintenance season.

13. RISK ANALYSIS AND MITIGATION

Risk	Level	Mitigation
Access to construction and condition records may be restricted, as such records may disclose deficiencies in past construction	High	The framework is presented as a forward-looking asset management instrument and not as an audit. Initial application to newly constructed corridors where records are complete.
Prospective validation requires years, as pavements deteriorate slowly	High	Retrospective validation against sections of known condition; selection of heavily trafficked corridors.
Field records may be incomplete or of variable quality	Moderate	The mechanistic core operates without the machine-learning layer; missing data widens the reported confidence interval rather than preventing computation.
Institutional procurement cycles are extended	Moderate	Initial engagement with concessionaires and consultants.
Reproduction of codal provisions is subject to copyright	Moderate	Computational procedures and numerical relationships are implemented; no protected expression is reproduced. Every output cites the governing clause. Formal permission is being sought.
Predictions may be relied upon in substitution for professional judgement	Moderate	Output is decision support requiring approval by a qualified engineer; confidence intervals and out-of-range warnings accompany every result; a complete computation trace is provided.

14. ETHICAL, LEGAL AND REGULATORY COMPLIANCE

- **Research ethics.** The work involves no human or animal subjects.
- **Intellectual property in codal provisions.** Copyright in Indian Standards and in Indian Roads Congress publications vests in the respective issuing bodies. The implementation reproduces computational procedures and numerical relationships only; no protected expression, table layout or clause text is reproduced. Formal permission for reproduction of extracts is being sought.
- **Confidentiality of client data.** Construction and condition records obtained from any authority or concessionaire will be treated as confidential, used solely for this research, and reported in anonymised form unless express written consent is obtained.
- **Data protection.** Where personal data is processed in any subsequent software deployment, the requirements of the Digital Personal Data Protection Act, 2023 will be complied with, including data minimisation.
- **Professional responsibility.** All output is advisory. The framework does not substitute for design by a qualified engineer, and every report carries an express statement to that effect.

15. CONCLUSION

Concrete pavements are presently designed upon assumptions that are not revisited, and maintained upon observation that arrives too late. The consequence is the simultaneous waste of material where conditions are favourable and the premature failure of segments where they are not. The mechanistic framework required to address this is already prescribed in IRC:58; what is absent is its probabilistic extension, its application forward in time, and its calibration against the records that construction already generates.

The proposed work supplies these elements. It does not displace the codal procedure but extends it, reducing exactly to the conventional computation when variability is absent. It requires no instrumentation, no proprietary software and no significant capital outlay, and a verified computational core has already been implemented.

The approval and guidance of the Supervisor is respectfully sought, together with such institutional support as may be considered appropriate for the acquisition of field records and the identification of a suitable corridor for demonstration.

(Signature of the Investigator)

(Signature of the Supervisor)

REFERENCES

- [1] IRC:58 — Guidelines for the Design of Plain Jointed Rigid Pavements for Highways. Indian Roads Congress, New Delhi.
- [2] IRC:SP:83 — Guidelines for Maintenance, Repair and Rehabilitation of Cement Concrete Pavements. Indian Roads Congress, New Delhi.
- [3] IRC:44 — Guidelines for Cement Concrete Mix Design for Pavements. Indian Roads Congress, New Delhi.
- [4] IS 456:2000 — Plain and Reinforced Concrete: Code of Practice. Bureau of Indian Standards, New Delhi.
- [5] Westergaard, H. M. (1926). Stresses in concrete pavements computed by theoretical analysis. *Public Roads*, 7(2), 25–35.
- [6] Westergaard, H. M. (1927). Analysis of stresses in concrete pavements due to variations of temperature. *Proceedings, Highway Research Board*, 6, 201–215.
- [7] Bradbury, R. D. (1938). *Reinforced Concrete Pavements*. Wire Reinforcement Institute, Washington, D.C.
- [8] Ioannides, A. M., Thompson, M. R., & Barenberg, E. J. (1985). Westergaard solutions reconsidered. *Transportation Research Record*, 1043, 13–23.
- [9] Miner, M. A. (1945). Cumulative damage in fatigue. *Journal of Applied Mechanics*, 12(3), A159–A164.
- [10] Portland Cement Association (1984). *Thickness Design for Concrete Highway and Street Pavements*. Skokie, Illinois.
- [11] AASHTO (1993). *Guide for Design of Pavement Structures*. Washington, D.C.
- [12] Ministry of Road Transport and Highways. *Specifications for Road and Bridge Works*. Indian Roads Congress, New Delhi.
- [13] National Highways Authority of India (2025). *Maintenance Manual*.